

# HW4

Після запуску всього застосунок надсилаємо 10 повідомлень.

Логи із кожного з екземплярів logging\_service

```
INFO: Started server process [31844]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8001 (Press CTRL+C to quit)
INFO: 127.0.0.1:55916 - "POST / HTTP/1.1" 200 OK
INFO: logging_service:RECEIVED: uuid='f705453b-6726-49c5-9d86-9fa00891a91e' msg='8'
INFO: 127.0.0.1:55922 - "POST / HTTP/1.1" 200 OK
```

```
INFO: Started server process [31911]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8002 (Press CTRL+C to quit)
INFO: logging_service:RECEIVED: uuid='80efe296-2352-440e-8c70-4d5b82117005' msg='2'
INFO: 127.0.0.1:40854 - "POST / HTTP/1.1" 200 OK
INFO: logging_service:RECEIVED: uuid='26f75990-5f7c-45c8-ad18-d71fa4e288cb' msg='5'
INFO: 127.0.0.1:51052 - "POST / HTTP/1.1" 200 OK
INFO: logging_service:RECEIVED: uuid='58b4c06b-41a6-4aa8-a1a9-bb62593c57eb' msg='6'
INFO: 127.0.0.1:51056 - "POST / HTTP/1.1" 200 OK
INFO: logging_service:RECEIVED: uuid='96751879-62aa-40e1-8162-c58de78553bb' msg='7'
INFO: 127.0.0.1:51066 - "POST / HTTP/1.1" 200 OK
INFO: logging_service:RECEIVED: uuid='52f2a1b9-a549-45ab-9dc4-eaecfa5e7015' msg='9'
INFO: 127.0.0.1:59428 - "POST / HTTP/1.1" 200 OK
INFO: logging_service:RECEIVED: uuid='9f11dcac-b1bc-4a28-aa82-d2ade2b39fa6' msg='10'
INFO: 127.0.0.1:59434 - "POST / HTTP/1.1" 200 OK
```

```
INFO: Started server process [32249]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8003 (Press CTRL+C to quit)
INFO: logging_service:Logs requested
INFO: 127.0.0.1:42048 - "GET / HTTP/1.1" 200 OK
INFO: logging_service:RECEIVED: uuid='b68a7efa-b5fe-44ef-bdc4-b73ae7f93dbc' msg='1'
INFO: 127.0.0.1:47942 - "POST / HTTP/1.1" 200 OK
INFO: logging_service:RECEIVED: uuid='5dad1863-9ed6-44c5-9517-099251b61cee' msg='3'
INFO: 127.0.0.1:57312 - "POST / HTTP/1.1" 200 OK
INFO: logging_service:RECEIVED: uuid='2cdd74f2-d5d8-4052-9577-20bce8866144' msg='4'
INFO: 127.0.0.1:42684 - "POST / HTTP/1.1" 200 OK
```

Те ж саме, але для messages\_client

```
INFO:hazelcast.connection:Authenticated with server Address(host=127.0.0.1, port=5801):b4c35ae2-cb6d-4948-9cc2-b8df2770c59f, server version: 5.3.8, local address: Address(host=127.0.0.1, port=43440)
INFO:hazelcast.client:Client started
INFO:messages_service:RECEIVED: 1
INFO:messages_service:RECEIVED: 3
INFO:messages_service:RECEIVED: 5
INFO:messages_service:RECEIVED: 7
INFO:messages_service:RECEIVED: 9
```

```
INFO:hazelcast.connection:Authenticated with server Address(host=127.0.0.1, port=5801):b4c35ae2-cb6d-4948-9cc2-b8df2770c59f, server version: 5.3.8, local address: Address(host=127.0.0.1, port=40856)
INFO:hazelcast.client:Client started
INFO:messages_service:RECEIVED: 2
INFO:messages_service:RECEIVED: 4
INFO:messages_service:RECEIVED: 6
INFO:messages_service:RECEIVED: 8
INFO:messages_service:RECEIVED: 10
```

Далі викличемо декілька разів GET, аби отримати повідомлення. Бачимо, що messages\_service дає на різні результати в залежності від того, який випадковим чином обрано екземпляр messages\_service.

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/
```

Server response

| Code | Details                                                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 200  | <div>Response body</div> <div><pre>{   "status": "ok",   "logging_service_messages": "2;1;9;7;10;6;8;5;4;3",   "messages_service_messages": "1;3;5;7;9" }</pre></div> |

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/
```

Server response

| Code | Details                                                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 200  | <div>Response body</div> <div><pre>{   "status": "ok",   "logging_service_messages": "2;1;9;7;10;6;8;5;4;3",   "messages_service_messages": "2;4;6;8;10" }</pre></div> |

Перевірка відмовостійкості черги повідомлень.

1.Запускаємо все, окрім 2 messages\_services.

2.Створюємо 10 повідомлень.

3.Вбиваємо процес із наймолодшим портом

```
(venv) sviatoslav:hw1$ ps aux | grep hazelcast | grep 5801
sviatos+  42251 15.5  1.9 10543732 279236 pts/4  Sl   15:54   0:15 /usr/bin/java -cp /usr/lib/hazelcast/lib:/usr
/lib/hazelcast/lib/*:/usr/lib/hazelcast/bin/user-lib:/usr/lib/hazelcast/bin/user-lib/* --add-modules java.se --a
dd-exports java.base/jdk.internal.ref=ALL-UNNAMED --add-opens java.base/java.lang=ALL-UNNAMED --add-opens java.b
ase/sun.nio.ch=ALL-UNNAMED --add-opens java.management/sun.management=ALL-UNNAMED --add-opens jdk.management/com
.sun.management.internal=ALL-UNNAMED -Dhazelcast.logging.type=log4j2 -Dlog4j.configurationFile=file:/usr/lib/haz
elcast/config/log4j2.properties -Dhazelcast.config=/usr/lib/hazelcast/config/hazelcast.xml -Djet.custom.lib.dir=
/usr/lib/hazelcast/custom-lib com.hazelcast.commandline.HazelcastServerCommandLine start --config=hazelcast/haze
lcast-messaging.xml --port=5801
(venv) sviatoslav:hw1$ kill -9 42251
(venv) sviatoslav:hw1$
```

4.Запускаємо 2 messages\_services.

Перевіряємо, чи є щось в messages\_services. Бачимо, що екземпляр на порті 8004 отримав всі повідомлення. Отже, жодного не було втрачено.

```
curl -X 'GET' \
'http://127.0.0.1:8004/' \
-H 'accept: application/json'
```

Request URL

http://127.0.0.1:8004/

Server response

| Code | Details |
|------|---------|
|------|---------|

|     |               |
|-----|---------------|
| 200 | Response body |
|-----|---------------|

```
{
  "status": "ok",
  "data": {
    "messages": [
      "1",
      "2",
      "3",
      "4",
      "5",
      "6",
      "7",
      "8",
      "9",
      "10"
    ]
  }
}
```

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8005/' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8005/
```

Server response

| Code | Details |
|------|---------|
|------|---------|

200

Response body

```
{
  "status": "ok",
  "data": {
    "messages": []
  }
}
```

Response headers